Derivative

Definition

f is said to be differentiable at a point $a \in \mathbb{R}$, $\Leftrightarrow$ f is defined on open interval I containing a, and:

$f'(a) := \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ exists, $f'(a)$ is call derivative of f at a

↓

remark: Can also written as: $\lim_{x\to a} \frac{f(x)-f(a)}{x-a}$

differentiability is a local property, like continuity.

Theorem

f is differentiable at a $\Leftrightarrow$ $\exists\, g: I \to \mathbb{R}$.

(1) g is continuous at a.

(2) $f(x) = g(x)(x-a) + f(a)$ $\forall\, x \in I$.

In this case. $f'(a) = g(a)$

proof: ① $g(x) = \begin{cases} \frac{f(x)-f(a)}{x-a}, & x \neq a. \\ f'(a), & x = a. \end{cases}$

Then $f(x) = (x-a)g(x) + f(a)$. $\forall x \in I$

$\lim_{x\to a} g(x) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a} = f'(a) = g(a) \Rightarrow$ g is cont. at a.

② $f(x) - f(a) = g(x)(x-a)$.

$f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a} = \lim_{x\to a} g(x) = g(a)$. $\Rightarrow$ f is differentiable at a, $f'(a) = g(a)$

Theorem: if f is differentiable at a, then f is continuous at a.

proof: f is diff. at a $\Rightarrow$ $f(x) - f(a) = g(x)(x-a)$, for some g cont. at a.

$\lim_{x\to a}[f(x) - f(a)] = \lim_{x\to a} g(x)(x-a) = \lim_{x\to a} g(a)(a-a) = 0$.

$\therefore \lim_{x\to a} f(x) = f(a)$. f is cont. at a.

Endpoint Derivative

$f_1'(\alpha) = \lim_{x\to \alpha^+} \frac{f(x)-f(\alpha)}{x-\alpha}$ $\quad f_1'(\beta) = \lim_{x\to \beta^-} \frac{f(x)-f(\beta)}{x-\beta}$ "one-sided limit"

Definition:

A function f is differentiable on I $\Leftrightarrow$ $f_1'(a) = \lim_{\substack{x\to a \\ x\in I}} \frac{f(x)-f(a)}{x-a}$ exists and finite. $\forall a \in I$.

A function f is continuously differentiable on I if f_1' exists & cont. on I.

Standard Function Space

$C^0(I) = \{ f: I \to \mathbb{R} :$ f is cont. on $I \}$

$C'(I) = \{ f: I \to \mathbb{R} :$ f' exist & cont. on $I \}$

$\Rightarrow C^0(I) \supseteq C'(I) \supseteq \cdots \supseteq C^{(n)}(I) \supseteq C^{(\infty)}(I)$

$C^{(n)}(I) = \{ f: I \to \mathbb{R} :$ $f^{(n)}$ exist & cont. on $I \}$

$C^{(\infty)}(I) = \{ f: I \to \mathbb{R} :$ $f^{(n)}$ exist & cont. on I for all $n \in \mathbb{N} \}$

$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}), & x \neq 0 \\ 0, & x = 0 \end{cases}$

(1) f is diff. on $\mathbb{R}$.

(2) f is not continuously diff at 0. $\Rightarrow f \notin C(\mathbb{R})$.

proof: (1). $\forall\, x \neq 0$. $f'(x) = 2x\sin(\frac{1}{x}) + x^2\cos(\frac{1}{x})\cdot(-\frac{1}{x^2})$

$= 2x\sin(\frac{1}{x}) - \cos(\frac{1}{x})$

f is diff. at any $x \neq 0$

$f'(0) := \lim_{x\to 0} \frac{f(x)-f(0)}{x-0}$

$= \lim_{x\to 0} \frac{f(x)}{x} = \lim_{x\to 0} x\sin(\frac{1}{x}) = 0$.

$\therefore f(x)$ is diff at all $x \in \mathbb{R} \Rightarrow f(x)$ is diff. on $\mathbb{R}$.

(2). $\lim_{x\to 0} f'(x) = f'(0)$?

$\lim_{x\to 0}(2x\sin(\frac{1}{x}) - \cos(\frac{1}{x})) = \lim_{x\to 0}(-\cos\frac{1}{x}) \Rightarrow$ DNE.

$\therefore f$ is not cont. diff. at 0

e.x. $f(x) = |x|$.

(1). f is diff. on $[0,1]$.

(2) f is diff. on $[-1,0]$

(3) f is not diff. on $[-1,1]$ $\quad f_1'(0) = \lim_{x\to 0+} \frac{f(x)-0}{x-0} = 1$

(1). on $(0,1]$ $f(x) = x$, $f'(x) = 1$ exists $\Rightarrow$ f is diff. on $[0,1]$

(2) on $[-1,0)$ $f(x) = -x$ $f'(x) = -1$ exists $f_1'(0) = \lim_{x\to 0-} \frac{f(x)-0}{x-0} = -1 \Rightarrow$ f is diff. on $[-1,0]$

(3). at $x=0$. $\lim_{x\to 0+} \frac{f(x)-0}{x-0} = 1$.

$\lim_{x\to 0-} \frac{f(x)-0}{x-0} = -1$. $\Rightarrow \lim_{x\to 0} \frac{f(x)-0}{x-0} = f'(x)$ DNE., f is not diff. at $x=0$

Hence f is not diff. on $[-1,1]$

Thm (Arithmetic): $f, g: I \to \mathbb{R}$. $a \in I$.

f, g are diff at a $\Rightarrow$ So is $f \pm g$ $f\cdot g$. $\alpha\cdot f$ ($\alpha \in \mathbb{R}$) at a

$\Rightarrow$ if $g(a) \neq 0$. so is $\frac{f}{g}$.

proof: $(f\cdot g)'(a) = \lim_{x\to a} \frac{(f\cdot g)(x)-(f\cdot g)(a)}{x-a}$.

(product). $= \lim_{x\to a} \frac{f(x)g(x) - f(a)g(a)}{x-a}$

$= \lim_{x\to a} \frac{f(x)g(x) - f(a)g(x) + f(a)g(x) - f(a)g(a)}{x-a}$

$= \lim_{x\to a} g(x)\frac{f(x)-f(a)}{x-a} + \lim_{x\to a} f(a)\frac{g(x)-g(a)}{x-a}$.

$= g(a)f'(a) + f(a)g'(a)$

Quotient: $(\frac{1}{f})'(a) = \lim_{x\to a} \frac{\frac{1}{g(x)} - \frac{1}{g(a)}}{x-a}$

$= \lim_{x\to a} \frac{g(a)-g(x)}{(x-a)(g(x)\cdot g(a))}$

$= -\frac{1}{g(a)^2} \lim_{x\to a} \frac{g(x)-g(a)}{x-a}$

$= -\frac{g'(a)}{g(a)^2}$

$\therefore (\frac{f}{g})'(a) = (f\cdot\frac{1}{g})'(a)$

Chain Rule

Suppose f is diff. at a and g is diff. at $f(a)$, then $g\circ f$ is diff. at a

$(g\circ f)'(a) = g'(f(a))\cdot f'(a)$

proof: Since f is diff. at a. $\exists\, F: f(x) = F(x)(x-a) + f(a)$.

g is diff. at $f(a)$ $\exists\, G: g(y) = G(y)(y - f(a)) + g(f(a))$

$g(f(x)) - g(f(a)) = G(f(x))(f(x) - f(a))$

Since f is diff. at a. f is cont. at a.

$(g\circ f)'(a) = \lim_{x\to a} \frac{g(f(x)) - g(f(a))}{x-a} = \lim_{x\to a} \frac{G(f(x))(f(x)-f(a))}{x-a} = G(f(a))F'(a) = g'(f(a))f'(a)$

Rolle's Theorem

Let $a < b$. if f is cont. on $[a,b]$, diff. on (a,b), $f(a) = f(b)$.

Then $\exists\, c \in (a,b)$ $f'(c) = 0$.

By Extreme Value Theorem

f has a maximum M and min m in $[a,b]$

if $m = M$. f is constant. $f(x) = 0$. $\forall\, x \in (a,b)$

if $m < M$. since $f(a) = f(b)$ $\exists\, c \in (a,b)$ $f(c) = m$ or M.

Suppose $f(c) = m$. Then $f(c+h) \geq f(c)$ $\forall$ sufficient small h.

Let $\{x_n\}$ be seq. such that $x_n \to c$ and $x_n > c$. $\Rightarrow \lim_{n\to\infty} \frac{f(x_n)-f(c)}{x_n - c} \geq 0$.

$\Vert$

$f'(c)$.

Let $\{y_n\}$ be seq. such that $y_n \to c$. and $y_n < c$

$\Rightarrow \lim_{n\to\infty} \frac{f(y_n)-f(c)}{y_n - c} \leq 0$.

$\Vert$

$f'(c)$.

$\Rightarrow f'(c) \geq 0$ and $f'(c) \leq 0 \Rightarrow f'(c) = 0$

Mean Value Theorem

if f is cont. on $[a,b]$ and diff on (a,b), then there is $c \in (a,b)$ such that

$f'(c) = \frac{f(b)-f(a)}{b-a}$, $\quad f'(c)(b-a) = f(b) - f(a)$

proof: Set Secant line

$l(x) = \frac{f(b)-f(a)}{b-a}(x-a) + f(a)$

Let $h(x) = f(x) - l(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a)$.

$h(a) = h(b) = 0 \Rightarrow \exists\, c \in (a,b)$. $h'(c) = 0$.

$\Rightarrow h'(c) = f'(c) - \frac{f(b)-f(a)}{b-a} = 0$.

$\Rightarrow f'(c) = \frac{f(b)-f(a)}{b-a}$

Monotone functions

Monotone functions

Let $E \subseteq \mathbb{R}$ be nonempty and $f: E \to \mathbb{R}$.

(i) The function f is *increasing* on E if $x_1 < x_2$ implies $f(x_1) \leq f(x_2)$; it is *strictly increasing* if the inequality is strict.

(ii) It is *decreasing* on E if $x_1 < x_2$ implies $f(x_1) \geq f(x_2)$; it is *strictly decreasing* if the inequality is strict.

(iii) It is *monotone* if it is either increasing or decreasing.

For example, x^2 is monotone on $[-1,0]$ and on $[0,1]$, but not on $[-1,1]$.

Derivative test for monotonicity

Suppose f is continuous on $[a,b]$ and differentiable on (a,b).

(i) If $f'(x) \geq 0$ on (a,b), then f is increasing on $[a,b]$; if $f'(x) > 0$, then f is strictly increasing.

(ii) If $f'(x) \leq 0$ on (a,b), then f is decreasing; if $f'(x) < 0$, then f is strictly decreasing.

(iii) If $f'(x) = 0$ on (a,b), then f is constant on $[a,b]$.

proof if $x_1 < x_2$ MVT: $\exists\, c \in (x_1, x_2)$ $f'(c) = \frac{f(x_2)-f(x_1)}{x_2 - x_1}$

$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$

since $x_2 - x_1 > 0$, $f(x_2) - f(x_1) > 0$ or < 0 only depends on $f'(x)$

$\Rightarrow$ $f'(x) > 0 \Rightarrow f(x_2) > f(x_1) \Rightarrow$ increasing

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Thm 4.18

Suppose: f is increasing on $[a,b]$

① if $c \in [a,b)$ then $f(c+)$ exists and $f(c) \leq f(c+) = \lim_{x\to c+} f(x)$.

② if $c \in (a,b]$ then $f(c-)$ exists and $f(c-) \leq f(c)$

③ if $c_1 < c < c_2$ then:

$f(c_1) \leq f(c-) \leq f(c) \leq f(c+) \leq f(c_2)$

proof (2): $E = f([a,c))$ $\quad s = \sup E = \sup_{x\in[a,c)} f(x)$.

$f(x) \leq f(c)$ for $x < c$ $\Rightarrow$ $s \leq f(c)$.

$\forall\, \varepsilon > 0$. Choose $x_0 < c$. $s - \varepsilon < f(x_0) \leq s$

$\Rightarrow$ For $x_0 < x < c$. $s - \varepsilon < f(x_0) \leq f(x) \leq s$

$|f(x) - s| < \varepsilon \Rightarrow$ $f(c-) = s \leq f(c)$

Theorem 4.19. if f is monotone on an interval I.

then the set $\{x \in I : f$ is not cont. at $c\}$ is at most Countable$\}$

4.21 Bernoulli theorem

Let $\alpha > 0$ and $x \in [-1, +\infty)$

(i) if $0 < \alpha \leq 1$ then $(1+x)^\alpha \leq 1 + \alpha x$.

(ii) if $\alpha \geq 1$ then $(1+x)^\alpha \geq 1 + \alpha x$

proof: using mean value theorem on t^α between 1 and $x+1$

$\exists\, c$: $\alpha c^{\alpha-1} = \frac{(x+1)^\alpha - 1^\alpha}{x} = (x+1)^\alpha - 1$.

$\alpha x c^{\alpha-1} = (x+1)^\alpha - 1$

if $\alpha \in (0,1)$. if $x \geq 0$. then $c \geq 1$. $c^{\alpha-1} \leq 1 \Rightarrow \alpha x c^{\alpha-1} \leq \alpha x$

if $-1 \leq x < 0$ then $0 < c < 1$. $c^{\alpha-1} \geq 1 \Rightarrow \alpha x c^{\alpha-1} \leq \alpha x$

$(x+1)^\alpha - 1 \leq \alpha x \Rightarrow (1+x)^\alpha \leq 1 + \alpha x$

if $\alpha \geq 1$ $\quad (1+x)^\alpha \geq 1 + \alpha x$

Thm 4.23. IVT For Derivatives

Suppose f is diff. on $[a,b]$ with one-sided derivatives at endpoints.

and $f'(a) \neq f'(b)$. if y_0 lies strictly between $f'(a)$ & $f'(b)$. then there

is an $x_0 \in (a,b)$ such that $f'(x_0) = y_0$

Assume $f'(a) < y_0 < f'(b)$ set $F(x) = f(x) - y_0 x$. EVT.: F attains minimum at some $x_0 \in [a,b]$

$F'(a) = f'(a) - y_0 < 0$ for all sufficiently small $h > 0$ we have $F(a+h) < F(a)$. So a is not minimum point.

Similarly. b is not minimum point. Hence $x_0 \in (a,b)$ Fermat theorem $F'(x_0) = f'(x_0) - y_0 = 0 \Rightarrow f'(x_0) = y_0$. $x_0 \in (a,b)$